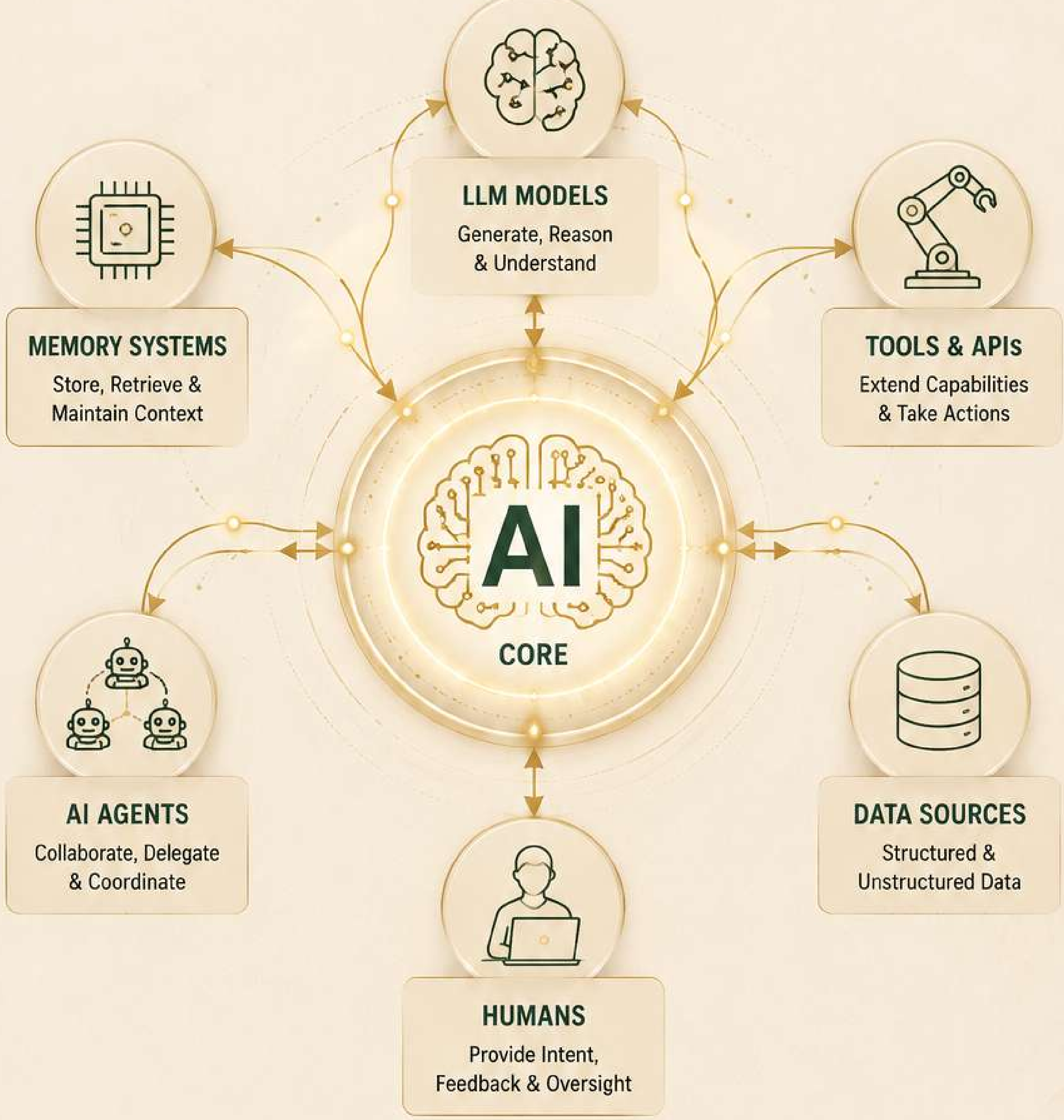
AI COMMUNICATION PROTOCOLS

HOW MODERN AI SYSTEMS EXCHANGE INFORMATION



EVERY INTELLIGENT SYSTEM DEPENDS ON COMMUNICATION.

VANSH BUDHIRAJA

THE COMMUNICATION ECOSYSTEM

AI SYSTEMS THRIVE WHEN THEY CAN CONNECT SEAMLESSLY

EVERY CONNECTION CREATES INTELLIGENCE

Protocols are the roads that allow data, tools, models and people to work together.

THE ROADS ARE PROTOCOLS

REST	MCP	WEBSOCKETS	SQL	JSON	EVENTS
Request / Response	Model ↔ Tool	Real-time Communication	Structured Data Access	Universal Data Format	Asynchronous Messaging

A CONNECTED ECOSYSTEM • SMARTER SYSTEMS • BETTER OUTCOMES

TYPES OF AI COMMUNICATION

Different ways AI systems connect and exchange information

AI CORE

1 HUMAN ↔ AI

Natural language conversations, prompts & feedback

2 AI ↔ TOOL

AI calls tools, plugins or functions to perform specific actions

3 AI ↔ API

API

Retrieve or send data via API endpoints (REST, GraphQL, etc.)

4 AI ↔ DATABASE

Read, write and query structured data for insights

5 AGENT ↔ AGENT

AI agents communicate, collaborate and coordinate to achieve goals

6 AI ↔ MEMORY

Store, retrieve and update context, knowledge and long-term memory

COMMUNICATION IS THE FOUNDATION OF INTELLIGENCE

The stronger the communication, the smarter the system.

ANATOMY OF AN AI REQUEST

> A complex request goes through multiple intelligent checkpoints to deliver accurate, relevant and actionable results.

Every request follows a journey before the final response is delivered.

01 USER
The user provides a question or instruction.

02 PROMPT
The request is structured and formatted for processing.

03 PLANNER
The system analyzes intent and creates an execution plan.

04 RETRIEVER
Relevant data, context and knowledge are retrieved.

05 TOOL
The required tool or API is called to perform the action.

06 LLM
The LLM reasons over the context and generates the output.

07 FINAL RESPONSE
The refined answer is sent back to the user.

THE JOURNEY IN ACTION

User Intent → System Understands → Context Collected → Tool Executed → AI Reasons → Answer Delivered

Intelligence is not just in the model.
It's in the journey that **connects everything.**

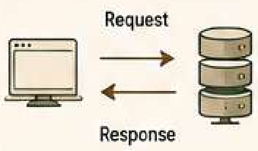
MODERN AI COMMUNICATION PROTOCOLS

Different protocols power different conversations.
Choose the right one for the right job.

REST APIs

Standard request–response communication over HTTP.

HOW IT WORKS



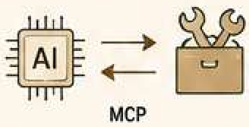
BEST FOR

Integrating with web services and external platforms.

MCP
(Model Context Protocol)

Standardized protocol for AI models to connect with tools and data sources.

HOW IT WORKS



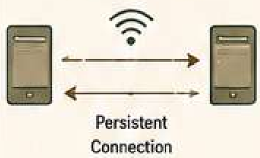
BEST FOR

AI-to-tool communication with rich context and capabilities.

WebSockets

Full-duplex, real-time communication over a single persistent connection.

HOW IT WORKS



BEST FOR

Real-time updates, streaming data and live interactions.

SQL

Structured query language for communicating with relational databases.

HOW IT WORKS

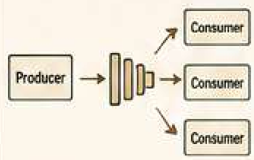
BEST FOR

Querying, updating and managing structured data.

Event Streaming

Asynchronous, event-driven communication using message brokers.

HOW IT WORKS



BEST FOR

Real-time pipelines, decoupled systems and scalable architectures.

AT A GLANCE COMPARISON

	REST APIs	MCP	WebSockets	SQL	Event Streaming
Communication Type	Request / Response	Contextual	Real-time Stream	Query / Response	Event-driven
Direction	Bi-directional	Bi-directional	Bi-directional	Bi-directional	One-to-Many
Connection	Stateless	Stateful (Context)	Persistent	Connection based	Persistent (Stream)
Data Format	JSON / XML	JSON	JSON / Binary	Tabular	JSON / Avro / Protobuf
Speed	Medium	High	Very High	Medium	Very High
Reliability	High (HTTP)	High	High	High (ACID)	Very High
Use Case	APIs, Integrations	AI ↔ Tools / Data	Live Apps, Real-time	Databases, Analytics	Pipelines, Microservices

No single protocol fits all.
The best AI systems use the right protocol at the right moment.

AI COMMUNICATION ARCHITECTURE

Well-structured communication makes AI systems more reliable, interpretable and impactful.

A layered communication architecture that powers intelligent, reliable and scalable AI systems.

USERS & APPLICATIONS

HTTPS / REST

1 AGENT ORCHESTRATION LAYER

Routes requests, manages sessions, chooses tools, and maintains the conversation state.

Decides what happens next and how.

MCP / REST

2 PLANNING & REASONING LAYER

Breaks down user intent, plans steps, and determines required information.

Thinks, plans and decomposes complex tasks.

MCP

3 MEMORY & CONTEXT LAYER

Stores conversation history, knowledge, preferences and long-term memory.

Provides context and continuity across sessions.

SQL / VECTOR

4 RETRIEVAL & DATA ACCESS LAYER

Retrieves relevant data from databases, vector stores, APIs and external systems.

Fetches accurate and relevant information.

REST / SQL

5 TOOL & SERVICE LAYER

Executes actions using tools, APIs, functions and external services.

Performs real-world actions and integrations.

API / MCP

6 MODEL & GENERATION LAYER

LLMs process the context and generate natural, accurate and helpful responses.

Generates the final response.

PROTOCOL KEY

- **REST** Standard HTTP-based communication
- **MCP** Connects models with tools and context
- **WebSockets** Real-time, bidirectional communication
- **SQL** Structured query and data access
- **Event Stream** Asynchronous messaging at scale

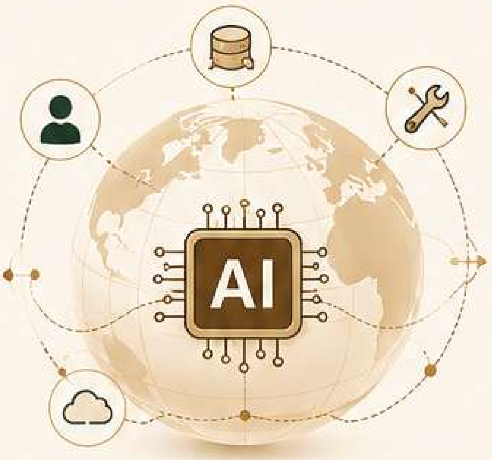
ENTERPRISE INFRASTRUCTURE

- Security & Governance
- Data Stores & Warehouses
- APIs & Third Party Services
- Monitoring & Observability
- Logging & Audit Trails

DESIGNED FOR INTELLIGENCE

- SECURE BY DESIGN
- MODULAR & FLEXIBLE
- SCALABLE BY NATURE
- RELIABLE & OBSERVABLE

END-TO-END
AI COMMUNICATION FLOW



From user intent to intelligent response — powered by the right protocols at every step.

01 USER INTENT
The user asks a question or gives a command.

Protocols: HTTPS / REST
Web / Mobile / Apps

02 INTENT UNDERSTANDING
The system understands the intent and extracts key information.

Protocols: MCP
Model ↔ Context

03 PROTOCOL SELECTION
The system chooses the best protocol to communicate with the right resource.

Protocols: Router Logic
Policy Engine

04 TOOL / DATA ACCESS
Relevant tools or data sources are accessed securely to gather the needed information.

Protocols: SQL / REST / API / MCP

05 CONTEXT & MEMORY
Information is enriched with context, history and long-term memory.

Protocols: Vector DB / Memory APIs

06 REASONING & GENERATION
The LLM reasons over the context and generates a meaningful response.

Protocols: LLM API (HTTPS / REST)

07 RESPONSE DELIVERY
The final response is delivered back to the user in real-time.

Protocols: WebSockets / HTTPS / SSE

USER → AI SYSTEM → VALUE DELIVERED

GREAT AI ISN'T JUST SMART.
IT'S CONNECTED.

KEY TAKEAWAYS

- **Right Protocol** — Use the right protocol for the right purpose.
- **Layered Design** — Modular layers make systems flexible, scalable and easier to evolve.
- **Secure by Default** — Security, governance and observability are built-in at every layer.
- **Real-time Ready** — Modern protocols enable real-time, bi-directional interactions.
- **Built for Scale** — A strong communication foundation unlocks reliability and enterprise scale.

Communication is the bridge between intelligence and impact.
Better protocols. Better systems. **Better outcomes.**